

Full length article

Research on surface defect imaging detection and traffic safety optimization of vehicle-mounted lidar wedge prisms

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ABSTRACT

Wedge prisms present significant technical challenges in optical detection due to their unique geometric structure. The acquisition of high-quality datasets remains a critical challenge in this field. This study proposes a novel method based on incident angle optimization and reflection area prediction to enhance imaging quality. The method ensures optimal component quality during manufacturing processes before LiDAR system integration. Using Mie scattering theory, we determined the optimal incident angle range of 20°–40°. Dual wavelength imaging demonstrated significant improvements across multiple performance indicators compared to single wavelength approaches. The established model relating wedge thickness to light reflection characteristics successfully predicted physical boundaries of two distinct reflection areas. Experimental validation confirmed the existence of interference reflection and geometric reflection mechanisms. The prediction error between the theoretical and actual image reflection areas remained within 20 pixels. Quantitative analysis reveals that micron-level optical defects significantly impact vehicle mounted LiDAR performance. This impact becomes particularly pronounced under heavy rainfall conditions, substantially reducing system reliability. This framework provides a foundation for improving LiDAR manufacturing quality control and enables future development of automated inspection systems.

1. Introduction

Wedge prisms serve as core components of high-precision optical devices in intelligent transportation systems. These components enable optical path modulation in vehicle LiDAR, zoom control in traffic monitoring cameras, and beam steering in tunnel lighting systems [1–3]. Recent advances demonstrate their expanding applications in polarization imaging and thermoelectric device detection [4–7].

Wedge prisms are crucial for mechanical LiDAR systems in L5 autonomous vehicles. These systems require sub-degree accuracy for obstacle detection at ranges exceeding 200 m and wide-angle scanning for pedestrian detection [8–11]. Environmental conditions such as heavy rainfall and dense fog introduce wavelength-selective attenuation that compromises wedge prism performance [12,13]. The inherent angular dispersion characteristics cause different wavelengths to refract

at varying angles, resulting in beam deviation and degraded ranging accuracy [14].

Surface quality directly impacts sensing performance. Vibrations from moving vehicles exacerbate microscopic defects on wedge prism surfaces [15,16]. Surface pits generate scattering effects ten times more intense than internal scattering [17,18]. This creates unwanted stray light that distorts point cloud data and compromises obstacle identification [19]. Such distortions pose serious safety risks and can trigger cascading failures in high-power laser systems [20]. Manufacturing stage defect detection is essential since damaged components require costly complete replacement [21].

Current imaging with wedge prisms faces significant challenges. The angle of the wedge causes uneven light refraction and complex scattering distributions [22]. Existing approaches inadequately consider light-path deflection, resulting in insufficient contrast for small defect

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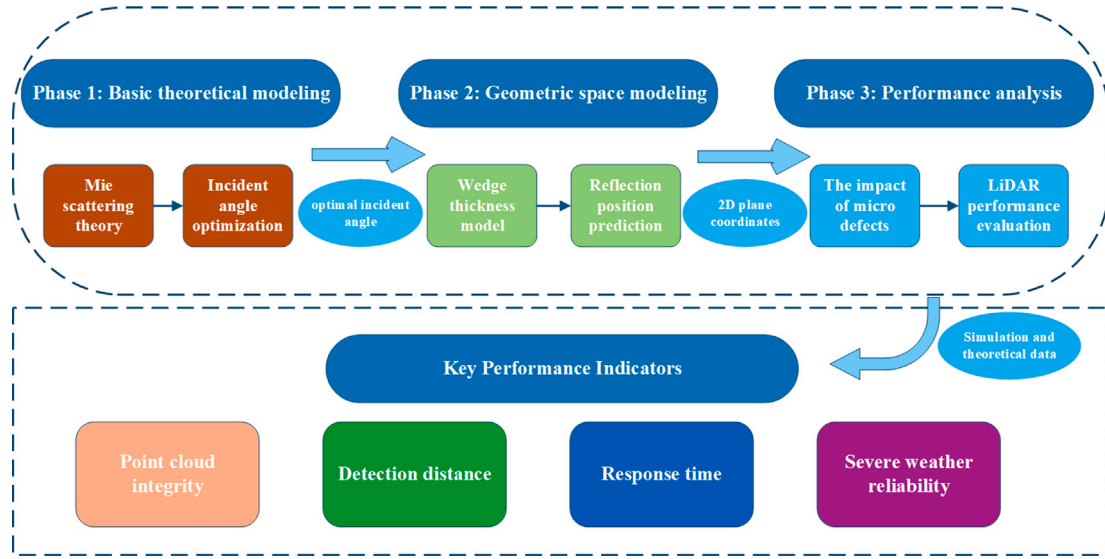


Fig. 1. Wedge prism imaging optimization method.

detection. Thickness gradients create an uneven spatial distribution of image quality [23].

While recent advances in finite difference time domain methods [24], illumination parameter optimization [25], image processing-based defect detection [26], and deep learning integration [27] show promise, most focus on planar components and lack integration with traffic engineering applications. Although autofocus systems [28] and multi-image fusion technology [29] can improve image quality, they require suitable shooting conditions and show limited effects in practical applications.

This study proposes an efficient wedge prism imaging optimization method based on incident angle optimization and reflection position prediction. First, we need to develop a relationship model between incident angle and contrast using Mie scattering theory to determine optimal illumination conditions. Then, we need to construct a wedge thickness model to predict the spatial distribution of interference and geometric reflection areas on imaging planes. Furthermore, the research aims to analyze the theoretical impact of optical component micro defects on vehicle mounted LiDAR performance, focusing on autonomous vehicle detection requirements for distant obstacles at high speeds and examining key performance indicators including point cloud integrity, detection distance, response time, and severe weather reliability. Fig. 1 illustrates the research workflow.

2. Model establishment

2.1. Scattering intensity calculation

2.1.1. Mie scattering theory

The characteristic size of pitting defects on wedge prism surfaces is comparable to the incident light wavelength. This places them in the Mie scattering regime where neither geometric optics nor Rayleigh scattering theories apply [30]. This framework enables quantitative analysis of scattered field variations with incident angle, defect size, and wavelength.

For surface pitting defects, the scattered field can be expressed as Eq. (1), where E_0 represents the incident wave amplitude, which depends on light source power, beam cross-sectional area, and medium properties. The terms N_{eln} and M_{oln} are vector spherical harmonics representing electric and magnetic field components respectively. These can be obtained from standard electromagnetic scattering theory tables [31]. The scattered field E_s determines the contrast between the defect area and

the background area.

$$E_s = E_0 \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (ia_n N_{eln} - b_n M_{oln}) \quad (1)$$

The scattering coefficients a_n and b_n determine the scattering intensity distribution and directional characteristics after the interaction between a light wave and a particle. These coefficients are expressed as Eq. (2).

$$\begin{aligned} a_n &= \frac{\psi_n(y)\psi'_n(x) - n\psi_n(x)\psi'_n(y)}{\xi_n(y)\psi'_n(x) - n\psi_n(x)\xi'_n(y)} \\ b_n &= \frac{n\psi_n(y)\psi'_n(x) - \psi_n(x)\psi'_n(y)}{n\xi_n(y)\psi'_n(x) - \psi_n(x)\xi'_n(y)} \end{aligned} \quad (2)$$

Where ψ_n and $\xi_n(z)$ are Riccati Bessel functions related to spherical Bessel and Hankel functions. The parameter n represents the complex refractive index ratio between the defect and surrounding medium, calculated as $n = n_2/n_1$, where n_2 is the refractive index of the defect material and n_1 is the refractive index of the surrounding medium. The size parameter x is defined as $x = 2\pi r/\lambda$, where r is the radius of the pitting defect and λ is the wavelength of incident light.

The a_n and b_n coefficients can be used to calculate the relationship between the scattered light intensity and the observation angle θ through Eq. (3).

$$I(\theta) = I_0 \cdot \frac{|S_1(\theta)|^2 + |S_2(\theta)|^2}{2k^2 L^2} \quad (3)$$

where I_0 is the incident light intensity, $S_1(\theta)$ and $S_2(\theta)$ are the scattering amplitude functions derived from the coefficients a_n and b_n , which can be calculated by Eq. (4), where $k = 2\pi/\lambda$ is the wave number, and L represents the observation distance from the scattering center. This formula can quantitatively predict the spatial distribution of scattered light from defects of specific size and optical properties. By analyzing this distribution, the optimal imaging angle to maximize the defect contrast can be determined.

$$\begin{aligned} S_1(\theta) &= \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \pi_n(\cos \theta) + b_n \tau_n(\cos \theta)] \\ S_2(\theta) &= \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \tau_n(\cos \theta) + b_n \pi_n(\cos \theta)] \end{aligned} \quad (4)$$

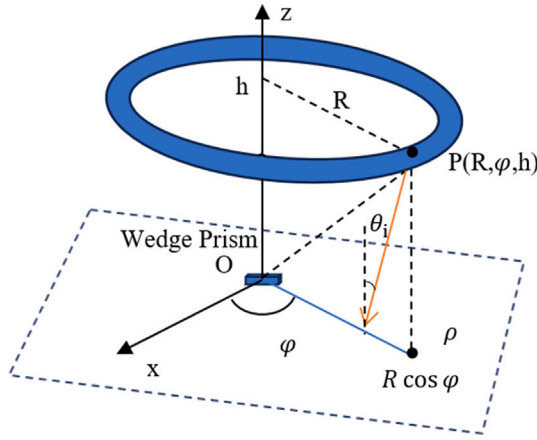


Fig. 2. Lighting space geometry.

where the angular distribution functions $\pi_n \cos \theta$ and $\tau_n \cos \theta$ describe scattered field changes with angle. The initial values of $\pi_n \cos \theta$ start from the first two terms, 0 and 1, and $\tau_1 \cos \theta = \cos \theta$.

Two key formulas can determine the angular scattering characteristics of surface pit defects and provide a theoretical basis for determining the optimal incident angle.

2.1.2. Annular light source modeling

To accurately describe the spatial geometric relationship between the annular light source and sample surface [32], a cylindrical coordinate system (ρ, φ, z) is established as shown in Fig. 2. The annular light source is positioned above the sample surface, with its central axis perpendicular to the sample plane.

Fig. 2 shows the cylindrical coordinate system (ρ, φ, z) , the sample surface with reference point O , the point $P(R, \varphi, h)$ on the light source, and the angle of incidence θ_i . R is the radius of the ring light, h is the perpendicular distance from the light source to the sample surface, and φ is the azimuth angle. $\cos \varphi$ is the projection of the Light source radius onto the incident plane that contains the normal surface and the incident ray.

Using these geometric parameters, Eq. (5) calculates the tangent value $\tan \theta_i$ of the incident angle from any point P of the annular light source to the sample surface, which directly represents the tilt

relationship between the incident light and the surface normal.

$$\tan \theta_i = \frac{R \cos(\varphi)}{h} \quad (5)$$

The intensity distribution of the light source follows a Gaussian profile described by Eq. (6), where I_{max} represents the maximum intensity and σ characterizes the distribution width.

$$I(\varphi, \rho) = I_{max} \exp\left(-\frac{\varphi^2}{2\sigma^2}\right) \quad (6)$$

The total scattering effect of the annular light source can be obtained by integrating over the light source area, as shown in Eq. (7), where W represents the effective width of the light source, θ changes with the position (ρ, φ) of each point on the ring light source, and it can be calculated using Eq. (3).

$$I_A = \int_0^{2\pi} \int_{R-\frac{W}{2}}^{R+\frac{W}{2}} I(\varphi, \rho) I(\theta) d\rho d\varphi \quad (7)$$

This section models the geometric characteristics of the ring light source in detail. By integrating over the full range of azimuth angles from 0 to 2π , the axial symmetry of the ring light source is taken into account, so that the total scattering effect of light reaching the sample surface at different incident angles from different positions of the ring light source can be accurately calculated.

2.1.3. Wedge angle compensation for incident angle

Wedge prisms have a significant effect on the propagation path of light, and for annular incident light, a comprehensive analysis of the angle compensation effect is required. With the continuous change in the azimuth direction, the actual compensation angle shows a dynamic change characteristic [33], reaching its minimum value of negative alpha at $\varphi = 0^\circ$ and its maximum value of positive alpha at $\varphi = 180^\circ$. Fig. 3 intuitively illustrates the principles of angle deflection and azimuth compensation.

The relationship between the actual incident angle θ_r between the light and the normal line of the wedge prism surface and the incident angle θ_a on the plane can be quantified by Eq. (8), where α is the wedge angle. Based on this compensation, the actual scattered light intensity is $I(\theta_r)$.

$$\theta_r = \theta \pm \alpha \cos(\varphi) \quad (8)$$

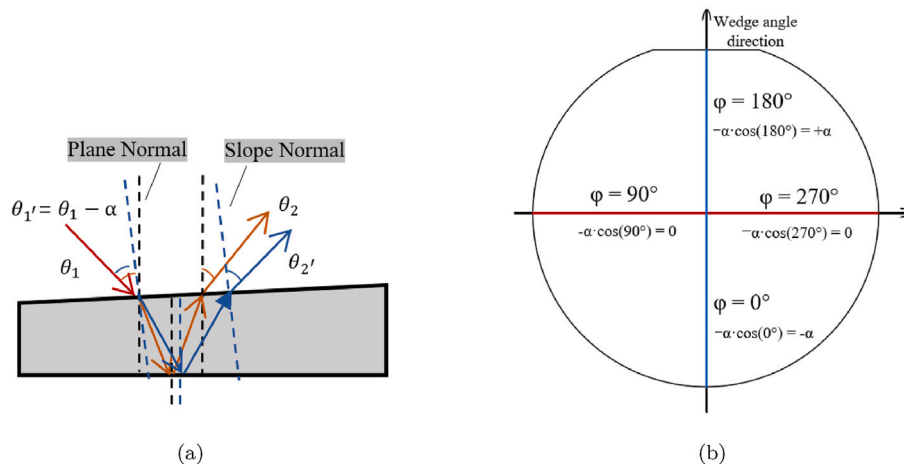


Fig. 3. Wedge angle compensation diagrams: (a) Angle deflection compensation, (b) Azimuth compensation.

2.2. Incident angle and contrast model

Based on the established scattered light intensity calculation method, the imaging contrast directly depends on the difference in scattering intensity between the defect and the background area. Therefore, the incident angle and contrast model can quantitatively analyze the difference in scattering intensity under different incident angles, thereby determining the optimal illumination angle for achieving maximum contrast. This quantitative relationship requires analyzing scattering intensity differences between pit region P versus background region B . Surface morphology changes in pit regions generate additional scattering [34], causing the pit region intensity to combine basic scattering with additional components as described by Eq. (9), while the background region exhibits only basic scattering intensity $I(\theta_r)$.

$$I_P(\theta_r) = I(\theta_r) + I_E(\theta_r) \quad (9)$$

$I_E(\theta_r)$ represents the additional scattered light intensity caused by the pit, which can be calculated using Eq. (10). The coefficient 0.3 is the scattering coefficient of the optical element, d and λ are the pit diameter and incident light wavelength, n_1 and n_2 are the atmospheric and material refractive indices, where the first four terms characterize the geometric and material properties of the pit, and the last two terms take into account the incident light conditions.

$$I_E(\theta_r) \approx 0.3 \cdot \left(\frac{d}{\lambda}\right)^2 \left(\frac{h}{d}\right) |n_1 - n_2|^2 I_A \cos(\theta_i) \quad (10)$$

The imaging contrast is related to the light intensity of the defect area and the background area. Eq. (11) shows the relationship between contrast and light intensity.

$$C = \frac{1.2\pi^2 R^2 \left(\frac{d}{\lambda}\right)^2 \left(\frac{h}{d}\right) |n_1 - n_2|^2 \cos(\theta_i)}{\lambda^2 \left(|S_1(\theta_r)|^2 + |S_2(\theta_r)|^2 \right) + 1.2\pi^2 R^2 \left(\frac{d}{\lambda}\right)^2 \left(\frac{h}{d}\right) |n_1 - n_2|^2 \cos(\theta_i)} \quad (11)$$

In order to facilitate the comparison of contrast performance under different conditions, all calculated contrast values are normalized to obtain the standardized contrast value C_n in Eq. (12).

$$C_n = \frac{C - C_{min}}{C_{max} - C_{min}} \quad (12)$$

C_{min} and C_{max} are the minimum and maximum contrast values observed under all measurement conditions, respectively. This normalization process ensures that all results are on a unified [0,1] scale, which is convenient for intuitive evaluation and threshold setting.

2.3. Optimal incident angle range

The wedge prism uses H-K9L optical glass material with high transparency in the visible to near-infrared band (400 nm–1100 nm). This ensures efficient illumination beam penetration through the prism surface. However, if a single white light is used, its dispersion characteristics will make it difficult to achieve uniform focusing in different areas of the wedge prism, affecting the accuracy of defect recognition. To address dispersion-induced focusing variations and wavelength-dependent Mie scattering limitations, the optical configuration incorporates a dispersion tube lens and beam splitter assembly for selective wavelength extraction. This choice is based on their complementary defect scattering properties, meaning that the 550 nm wavelength is particularly sensitive to tiny surface scratches and pits, while the 630 nm wavelength is more conducive to detecting subsurface collapse defects [35,36]. This dual wavelength approach optimizes focus under the wedge structure. It balances dispersion effects and achieves better image sharpness across different thickness areas, significantly improving overall defect imaging quality.

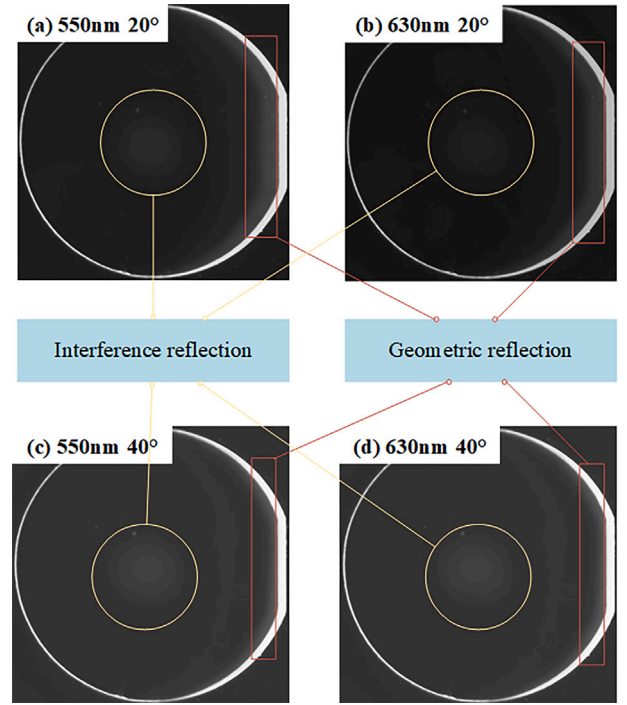


Fig. 4. Center and edge area reflective properties: (a)–(d) show the reflection of 550 nm and 630 nm wavelengths at 20° and 40°.

When determining the optimal incident angle for each wavelength light source, the effective angle range must satisfy the average contrast constraint of 0.6 and 0.95 in Eq. (13). This ensures sufficient contrast for detecting tiny defects while minimizing stray light interference.

$$\theta_\lambda = \{\theta, \varphi | C_m(\theta_r, \varphi) \in [0.6, 0.95]\} \quad (13)$$

The final effective detection angle range is determined by Eq. (14). This ensures both wavelengths achieve ideal detection at the same incident angle, enabling effective dual-wavelength information fusion. The wedge angle compensation creates asymmetric incident angle distribution on the left and right sides, providing additional scattering information that enhances pitting defect identification.

$$\theta_f = \theta_{\lambda_1} \cap \theta_{\lambda_2} \quad (14)$$

2.4. Wedge thickness and light reflection feature model

2.4.1. Interference and geometric reflection theory

The wedge prism surface shows reflection in the central detection area A and the wedge edge detection area B, with completely different optical behaviors. Born et al. [37] found that interference reflection light intensity is periodic and changes significantly with wavelength, while geometric reflection light intensity is evenly distributed and follows geometric optics laws. Kujawinska et al. [38] confirmed that the reflection intensity ratio has a nonlinear relationship with interference area and a linear relationship with geometric reflection area. Fig. 4 shows wavelength-sensitive interference fringes in the central prism area, while the wedge edge area exhibits geometric reflection characteristics with reflection intensity changing significantly with incident angle.

Interference reflection can be analyzed and understood using the optical path difference. Eq. (15) provides the conditions for the formation of interference reflection, where t is the thickness of the wedge prism at a certain position, θ_i is the refraction angle, m is the interference order, which indicates how many times the optical path difference equals the wavelength, and L_c is the coherent light source, which represents the maximum optical path difference at which light waves can interfere

stably. In the central area, the reflected light from the front and rear surfaces produces an optical path difference, which forms interference under the limitation of the coherence length of the light source. The refraction angle and coherence length are calculated using Eqs. (16) and (17) respectively.

$$\Delta = 2td \cos(\theta_t) = m\lambda < L_c \quad (15)$$

$$\theta_t = \arcsin \left[\frac{n_1}{n_2} \cdot \sin(\theta_r) \right] \quad (16)$$

$$L_c = \frac{\lambda^2}{\Delta\lambda} \quad (17)$$

When Eq. (18) is satisfied, the bright interference fringes of the same wavelength overlap at a specific position, resulting in a significant increase in brightness, forming a reflection phenomenon in the central area.

$$m_1 \lambda_1 \approx m_2 \lambda_2 \quad (18)$$

For a wedge prism, the expression for the thickness $t(z)$ varying with position z is given by Eq. (19).

$$t(z) = t_0 + z \cdot \tan(\alpha) \quad (19)$$

Where t_0 is the initial thickness and α is the wedge angle.

Based on the optical path difference theory, the effective thickness range for the reflection phenomenon in the central area is as shown in Eq. (20).

$$t_{min} \leq t \leq t_{max} \quad (20)$$

According to the research of Quirrenbach [39], the relationship between the visibility of interference fringes and thickness can be expressed as Eq. (21).

$$\begin{aligned} t_{min} &= \frac{m_{min} \cdot \lambda}{n_2} \\ t_{max} &= \frac{m_{max} \cdot \lambda}{n_2} \end{aligned} \quad (21)$$

Where m_{min} and m_{max} represent the minimum and maximum interference orders that can produce visible interference fringes.

Combining the wedge prism thickness function and the thickness condition, the position range of the central reflection area $[z_1, z_2]$ can be calculated as shown in Eq. (22).

$$\begin{aligned} z_1 &= \frac{t_{min} - t_0}{\tan(\alpha)} \\ z_2 &= \frac{t_{max} - t_0}{\tan(\alpha)} \end{aligned} \quad (22)$$

The reflection area at the wedge angle position is $[z_3, z_{max}]$, where z_3 can be determined by the critical value of Eq. (23).

$$z_3 = \frac{t_1 - t_c}{\tan(\alpha)} \quad (23)$$

where t_1 represents the true maximum thickness, and t_c is the critical thickness that exceeds the interference effect, which is usually greater than t_{max} .

The geometric reflection region starts at the critical thickness t_c . The reflection region is formed because part of the incident light is refracted by the prism and reflected by the bottom of the prism. When it reaches the right side, it is internally reflected again. The internally reflected light reaches the upper surface of the prism again and refracts into the atmosphere, thus forming an observable reflection region.

The critical thickness t_c can be determined by tracing the light path and geometric relationships. When the incident light enters the prism through refraction, its propagation path is determined by the refraction angle θ_r .

The light must be able to reach the right side and reflect back to the upper surface to form a geometric reflection. The critical thickness t_c that can produce reflection is calculated using Eq. (24).

$$t_c = t_1 \cdot [1 - \tan(\theta_1) \cdot \tan(\alpha)] \quad (24)$$

where θ_1 is the complementary angle to the refraction angle t_c .

2.4.2. 2D mapping of reflection areas

The above model is based on the thickness variation of the wedge prism along a specific direction. In practical applications, we need to map the thickness interval to a two-dimensional image plane. According to the orthogonal transformation method of Zamboj et al. [40], the distribution of the thickness of the wedge prism on the two-dimensional plane can be expressed by Eq. (25).

$$t(x, y) = t_0 + [x \cos(\varphi) + y \sin(\varphi)] \tan(\alpha) \quad (25)$$

Where (x, y) is the physical plane coordinate relative to the center point of the prism, in millimeters, and is constrained by the prism diameter D .

For the thickness ranges $[t_{min}, t_{max}]$ and $[t_c, t_1]$ of the interference and geometric reflection regions, the corresponding two-dimensional plane reflection region R satisfies Eq. (26).

$$R = \left\{ (x, y) \mid (t_{min} \leq t(x, y) \leq t_{max}) \vee (t_c \leq t(x, y) \leq t_1), x^2 + y^2 \leq \left(\frac{D}{2}\right)^2 \right\} \quad (26)$$

In actual image systems, the two-dimensional reflection area needs to be further mapped to the image coordinate system. Assuming that the image size is $M \times M$, the center point is $(i_0, j_0) = (M/2, M/2)$, and the pixel size is p , the conversion relationship between physical coordinates and pixel coordinates is expressed by Eq. (27).

$$\begin{aligned} i &= i_0 + \frac{x}{p} \\ j &= j_0 + \frac{y}{p} \end{aligned} \quad (27)$$

The corresponding relationship between thickness and pixel coordinates in Eq. (28) is obtained.

$$t(i, j) = t_0 + [(i - i_0)p \cos(\varphi) + (j - j_0)p \sin(\varphi)] \tan(\alpha) \quad (28)$$

This mapping relationship can accurately predict the position of the reflective area in the image, thereby providing physical constraints for subsequent image processing and defect detection, and improving the accuracy and robustness of the algorithm.

3. Experimental validation

3.1. Experimental system setup

The detection platform uses a ring-shaped white light LED source with an inner diameter of 60 mm and an outer diameter of 80 mm. This integrates a dispersion tube lens and beam splitter system for dual wavelength generation. The dispersion lens provides controlled chromatic dispersion to separate constituent wavelengths. Interference filters select 550nm and 630nm wavelengths with ± 10 nm spectral half width, while a beam splitter directs these filtered wavelengths toward the wedge prism specimen. The imaging system uses an industrial camera with 2048×2048 resolution and 60 fps acquisition rate, paired with a $5\times$ microscope objective with 0.15 numerical aperture. Image quality optimization includes automatic exposure control, multi-frame averaging, and real-time dark field correction. The optical system is installed on a clean bench with temperature stability maintained at 23°C. Fig. 5 shows the experimental setup.

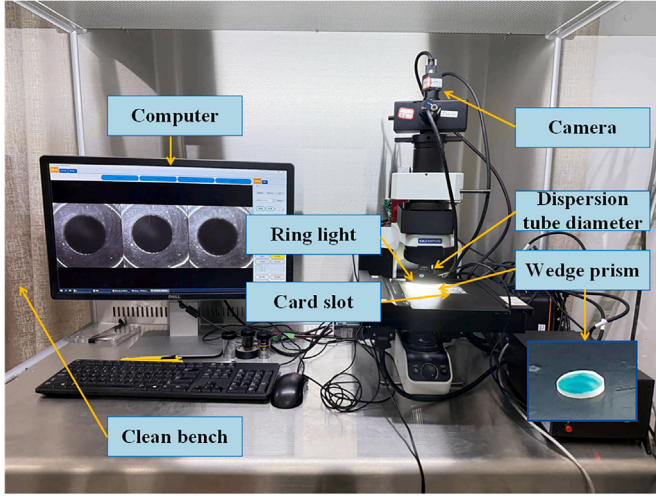


Fig. 5. Experimental equipment.

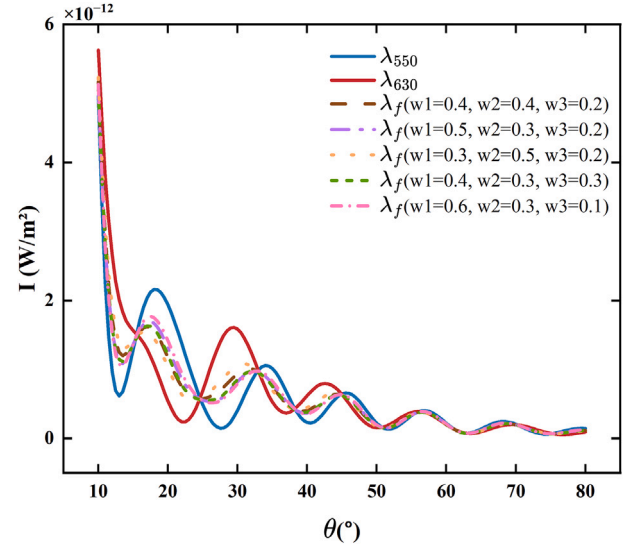


Fig. 6. Scattering intensity distribution.

3.2. Scattering characteristic validation

Due to the 5.5° wedge angle, incident light produces different actual incident angles at different azimuths. The left light path's actual incident angle increases by about 5.5° , while the right light path's actual incident angle decreases by about 5.5° . This asymmetric light path distribution creates asymmetric scattering intensity distribution [41]. Large scattering intensity differences between left and right sides increase imaging inhomogeneity, making uniform defect detection threshold setting difficult and increasing false detection and missed detection rates.

To verify the wedge prism angle compensation mechanism's effectiveness and its impact on imaging quality, the scattering intensity distribution at different angles and wavelengths was quantitatively measured. The incident angle measurement range was 10° – 80° with 2° intervals for 36 measurement points. To study the azimuth angle influence on scattering characteristics, the experiment measured the full circumference range from 0° – 360° with 10° intervals for 36 azimuth directions, achieving full angle characterization of scattering characteristics. Using dual wavelength light sources of 550 nm and 630 nm simultaneously achieves scattering characteristic fusion [42]. This fusion effect balances each wavelength's advantages and disadvantages while reducing wavelength dependence, obtaining more stable and representative scattering characteristic performance. The dual wavelength fusion method uses Eq. (29).

$$I_f(\rho, \varphi) = w_1 I_{550}(\rho, \varphi) + w_2 I_{630}(\rho, \varphi) + w_3 \left[\frac{I_{550}(\rho, \varphi)}{I_{630}(\rho, \varphi)} \right] \quad (29)$$

where $I_{550}(\rho, \varphi)$ and $I_{630}(\rho, \varphi)$ represent the scattering intensities at position (ρ, φ) under 550 nm and 630 nm wavelengths, respectively. The coefficients w_1 , w_2 , and w_3 are weight factors for the 550 nm scattering intensity, the 630 nm scattering intensity, and the ratio term, respectively, with their sum constrained to 1. This fusion method utilizes different scattering characteristics of the two wavelengths and uses the ratio term to compensate for geometric changes and anisotropic scattering introduced by the wedge structure, enhancing stability and representativeness of the final scattering characteristics.

To achieve optimal fusion performance, weight values are determined by analyzing scattering intensity distribution curves. Fig. 6 shows scattering intensity distribution curves of single 550 nm and 630 nm wavelengths, as well as fused wavelength curves from different weight coefficient combinations.

As the incident angle increases, the overall scattering intensity shows a downward trend. Dual wavelength fusion effectively mitigates extreme

peaks of single wavelengths. The weight combination of $w_1 = 0.4$, $w_2 = 0.4$, and $w_3 = 0.2$ provides moderate intensity levels, ensuring signals are neither too strong and distorted nor too weak and difficult to measure. This specific parameter set was used to calculate fused wavelength scattering intensity for subsequent performance evaluation. The interaction term $w_3 = 0.2$ avoids excessive inhibition of scattering features while maintaining signal smoothness, ensuring robust performance across the entire detection range.

To comprehensively evaluate system performance, this study introduced two key evaluation indicators: scattering uniformity and signal-to-noise ratio (SNR), compared with single-wavelength imaging performance. Scattering uniformity is defined as the degree of consistency in the scattering intensity distribution within the imaging area, expressed as 1 minus the percentage ratio of scattering intensity standard deviation to average value. Signal-to-noise ratio is defined as the ratio of the scattering intensity difference between the defect area and the background area to background scattering intensity standard deviation, directly related to the system's ability to detect tiny defects. Fig. 7 shows the relationship between scattering uniformity, signal-to noise ratio (SNR) and incident angle.

Table 1 compares the scattering characteristics of single wavelength and fusion wavelength at different incident angles, where θ_L and θ_R represent the actual incident angle after the incident angle θ is compensated by the left and right wedge angles, and $\Delta I(\%)$ is the difference in scattering intensity. The scattering intensity data of the fused wavelengths are selected to calculate $\Delta I(\%)$.

Data analysis shows that, compared with single wavelength, the fusion wavelength performs excellently in terms of scattering characteristics. The average scattering uniformity increases from 0.626 and 0.548 for the single wavelength to 0.744, and the average signal-to-noise ratio increases from 13.1 dB decibels and 9.3 dB decibels to 15.6 dB decibels. The difference in scattering intensity of the fusion wavelength is controlled within 30 %, and even reduced to less than 15 % at a relatively large incident angle, indicating that the detection stability and reliability of the system under various angular conditions are quite good.

3.3. Contrast evaluation validation

This experiment uses a standard wedge prism sample with a refractive index of 1.52, and the test object includes standard pit defects with diameters ranging from $3 \mu\text{m}$ to $8 \mu\text{m}$. The system performs contrast analysis through a sliding window of 45×45 pixels and calculates the

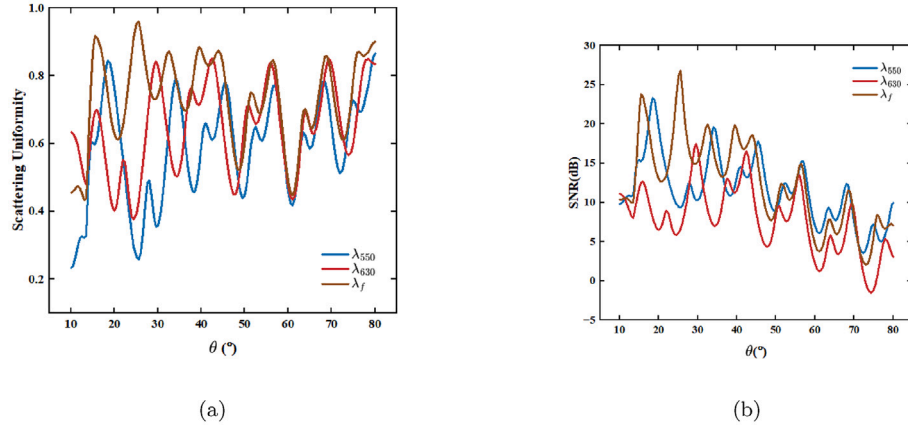


Fig. 7. Scattering intensity evaluation index: (a) Scattering uniformity, (b) Signal-to-noise ratio.

Table 1
Scattering evaluation index.

θ	θ_L	θ_R	$\Delta I(\%)$	Uniformity			SNR (dB)		
				λ_{550}	λ_{630}	λ_f	λ_{550}	λ_{630}	λ_f
20°	14.5°	25.5°	25.3	0.76	0.40	0.63	19.69	6.52	13.07
30°	24.5°	35.5°	24.5	0.36	0.83	0.74	10.26	16.69	15.23
40°	34.5°	45.5°	19.6	0.60	0.73	0.88	13.15	11.51	19.38
50°	44.5°	55.5°	16.4	0.45	0.66	0.6	8.94	8.11	8.95
60°	54.5°	65.5°	12.2	0.46	0.47	0.49	6.96	2.16	4.71

scattering intensity ratio between the pit area and the background area according to Eq. (13).

Four main requirements are set for performance evaluation: the contrast value (CR) must be maintained between 0.6 and 0.95; the coefficient of variation (CV) of the scattering intensity stability is required to be less than 10 %; the ROC curve area reflecting the comprehensive detection performance of the system must be above 0.8; and the minimum detectable defect size of the system is 3 μm . ROC curves can be generated by applying various contrast thresholds to the predicted scattering intensity distributions derived from your Mie scattering model, which calculates theoretical signal differences between 3 μm defects and background regions.

Through theoretical calculations, the effective dual-wavelength and fusion wavelength contrast threshold angle ranges are determined, as shown in Fig. 8.

From Fig. 8, it can be observed that the three wavelengths satisfy the contrast threshold in the angle ranges of [20°, 60°], [10°, 55°], and [10°, 40°], respectively. Considering the wedge-shaped asymmetric dual optical path incidence and the performance characteristics of these three wavelengths, the experiment determined that the optimal working angle range is 20°–40°, within which all three wavelengths can maintain stable high contrast performance.

Fig. 9 shows the performance comparison of 550 nm, 630 nm and fused wavelengths at multiple incident angles.

Table 2 summarizes the comparison of multiple evaluation indicators at optimal incident angles for the three wavelengths.

Data analysis shows that the fusion wavelength has significant advantages over single wavelength solutions: contrast can be increased by up to 25.0 %, coefficient of variation reduced by 41.3 %, and AUC increased by 11.5 %. The average contrast, coefficient of variation, and AUC of the fusion wavelength are all better than those of the single wavelength solution, which fully proves that this method improves the imaging quality of wedge prisms.

To verify the robustness of this method, we collected data from 50 standard wedge prism samples at 30° and 60° incident angles. Fig. 10

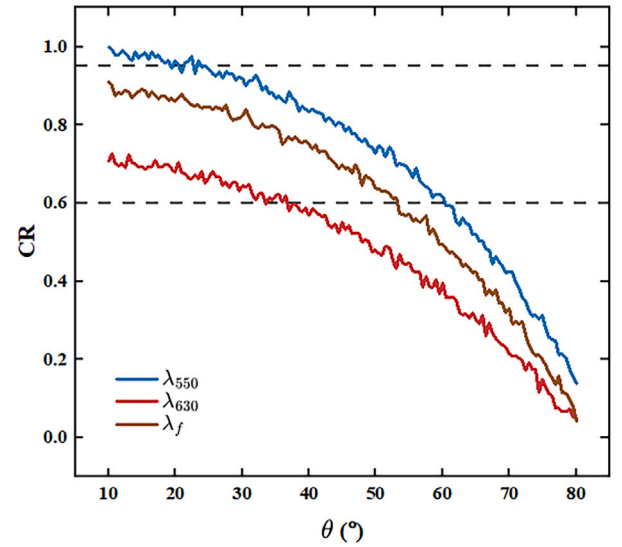


Fig. 8. Contrast threshold range.

shows the high-resolution imaging results of the same defective area of a wedge prism under three wavelength light sources, taken at the optimal incident angle range of 30° and 60° outside the area.

As shown in Fig. 10, the pit defect edges in the fusion wavelength image at 30° incidence angle are clearer, the contrast is higher, the background is more uniform, and the separation between defects provides a more reliable image basis for subsequent defect identification and measurement.

Based on Mie scattering theory, a quantitative relationship model between incident angle and contrast is established. This model accurately determines optimal illumination incident angle ranges for wedge prisms of different specifications and inclinations through a time offline calculation. The model uses Mie scattering equations to iteratively solve problems with low computational complexity, usually completed within seconds. Since optical calculations are completed in advance, no additional computational burden exists in subsequent detection processes, ensuring real-time performance and processing efficiency of the defect detection system.

3.4. Optical reflection area verification

This section explores the relationship between wedge thickness and interference-reflection area through the established dual-wavelength

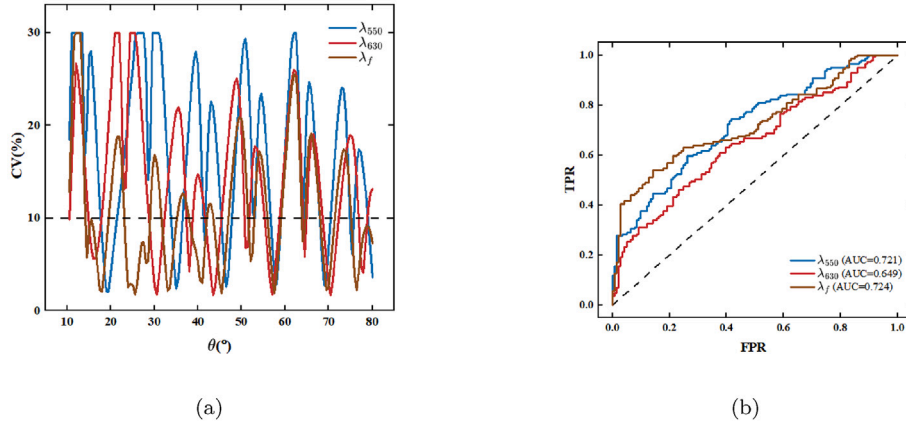


Fig. 9. Contrast evaluation index: (a) Coefficient of variation curve, (b) ROC curve.

Table 2
Optimized optical measurement results.

Parameter	Wavelength scheme	20°	30°	40°
CR	λ_{550}	0.94	0.63	0.88
	λ_{630}	0.92	0.60	0.83
	λ_f	0.83	0.52	0.76
CV(%)	λ_{550}	4.01	23.4	12.39
	λ_{630}	30.00	2.67	16.83
	λ_f	26.56	14.70	5.25
AUC	λ_{550}		0.721	
	λ_{630}		0.649	
	λ_f		0.724	

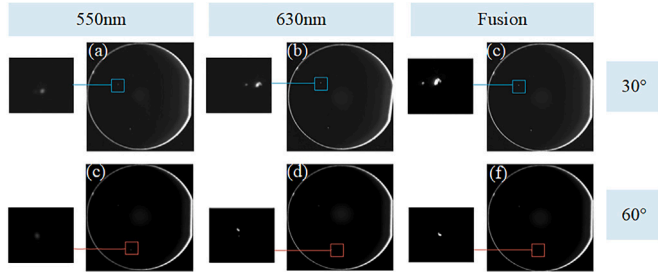


Fig. 10. Imaging contrast test experiment at different incident angles: (a)–(f) show the imaging effects of three wavelengths and two angles.

light source interference clarity calculation model. In the experiment, in order to calculate the thickness information of the interference reflection area, two wavelengths of light sources, 550 nm and 630 nm, were selected, and images of different batches of wedge prism samples were collected using a high-resolution camera. The thinnest and thickest parts of the wedge prism samples are 250 μm and 450 μm , respectively. A mathematical model including factors such as incident angle change, refractive index, and coherence length was established to calculate the interference order (m) and interference quality at different thicknesses. Interference quality is defined as the degree of deviation of the interference order from the nearest integer, that is, $Q = 1 - |m - \text{round}(m)|$, whose value is between 0 and 1, and the interference quality reaches its highest when the m value is an integer. The interference effect is shown in Fig. 11.

Fig. 11(a) shows linear growth of m values for two wavelengths with wedge thickness, confirming that the optical path difference is proportional to thickness. Shorter wavelength has a larger slope due to higher interference orders at the same thickness. Clustering occurs at 300 μm to 350 μm thickness range. Fig. 11(b) shows periodic interference quality fluctuation with thickness, with maximum values near 1 indicating clear interference patterns. This periodicity occurs as m_1 values approach integers with increasing thickness. Maximum interference quality occurs when m values are closest to integers, where both wavelengths form good interference and strong interference reflection phenomena occur.

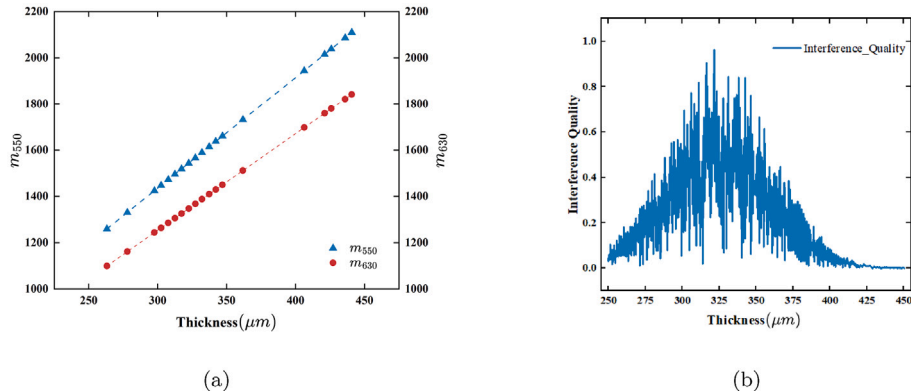


Fig. 11. Interference effect: (a) Interference level distribution, (b) Interference quality.

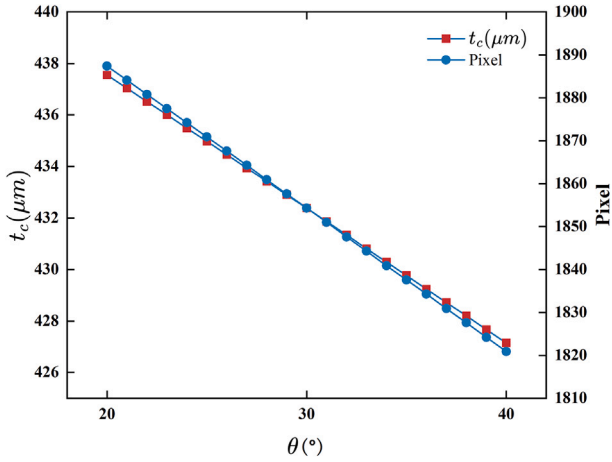


Fig. 12. Correspondence between thickness and pixel value.

Geometric reflection appears beyond critical thickness t_c . Determining t_c is key to verifying geometric reflection theory. In order to clearly show the distribution characteristics of the geometric reflection area, a critical thickness visualization scheme involving refraction supplementary angle, wedge prism geometric parameters, and wedge angle is designed. Fig. 12 shows the relationship between the thickness and pixel value of the wedge prism when geometric reflection occurs at different incident angles.

Through experimental observation, it is found that when light enters the optical system from different incident angles, the structural parameter t_c shows regular changes. At 20° incidence, the maximum value of t_c is measured to be 437.55 μm . As the incident angle increases, the t_c value continues to decrease, reaching a minimum value of 427.15 μm at 40°. Within the entire test range, the optical system exhibits a highly linear response characteristic.

Based on the pixel conversion formula, pixel coordinates of the thickness position where the wedge prism reflects are obtained and compared with experimental images. The wedge angle is uniformly placed to the right to facilitate pixel counting. Fig. 13 shows the comparison results.

Fig. 13 shows that pixel values of the geometric reflection theoretical area and actual image area are within 20 pixels, corresponding to a physical deviation of about 8 μm after imaging system calibration. The predicted pixel coordinate interval of the central interference reflection area is [750,1120]. Although the reflection area prediction model shows high accuracy, measurement errors exist due to lens distortion, noise limitations, and non-ideal placement factors such as prism offset or tilt in the tooling tray affecting image edge alignment.

Model prediction errors do not cause overall defects missing in edge areas, but mainly affect defect classification accuracy and extremely subtle feature recognition. Future research will adopt higher resolution hardware systems and develop more accurate prism positioning and calibration mechanisms to improve model accuracy and application reliability.

Results demonstrate that the established optical model has accurate prediction capabilities and can distinguish the physical boundaries of

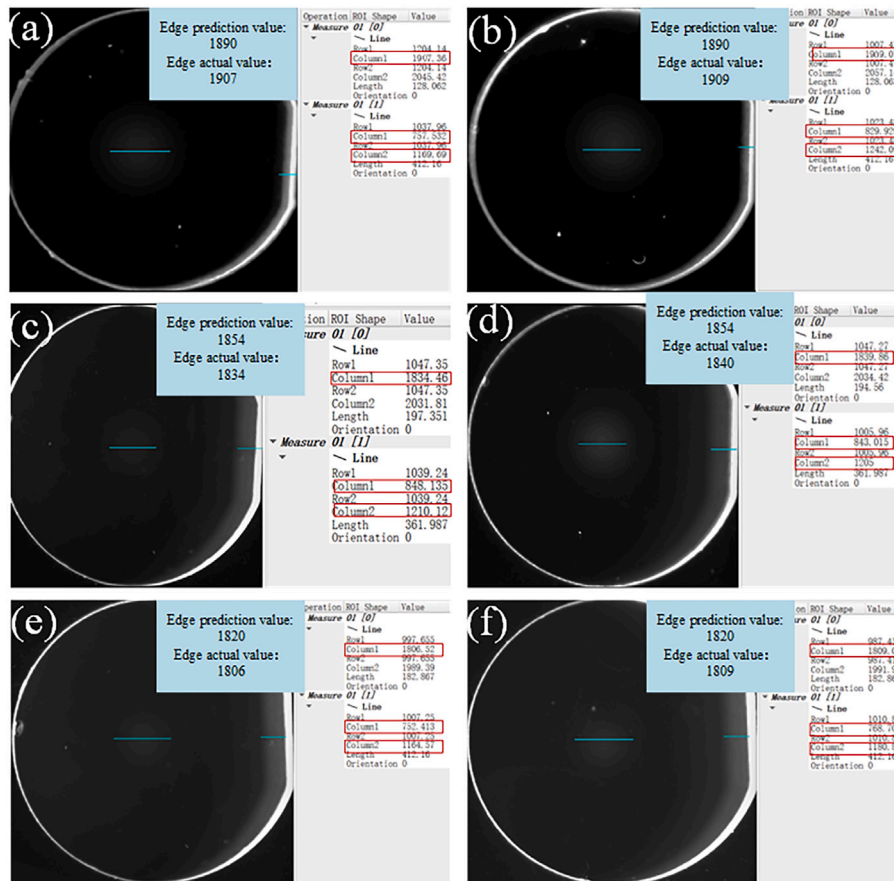


Fig. 13. Reflection theory area and real area test: (a)–(f) show pixel representations of the theoretically predicted boundaries of the reflective regions under different imaging conditions.

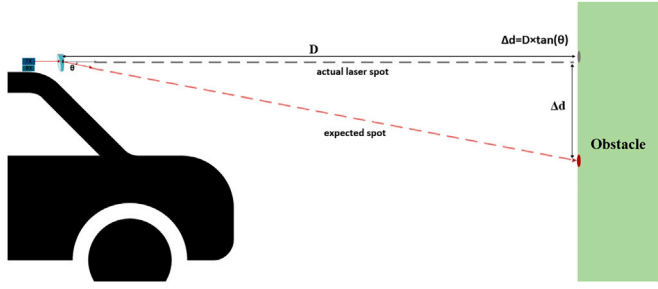


Fig. 14. Conceptual diagram of LiDAR ranging showing beam deviation caused by surface defects.

two reflection mechanisms. This verifies the theoretical model reliability of reflection mechanism transformation in different thickness regions. Accurate prediction of reflection region physical boundaries provides key physical prior knowledge and precise ROI guidance for subsequent image processing, machine learning, or deep learning algorithms. By concentrating computing resources on key areas, this method significantly reduces algorithm computational complexity and accelerates real-time defect detection processing speed.

4. Discussion

This section presents a theoretical analysis of how micron-level surface defects on wedge prisms impact vehicle-mounted LiDAR performance and traffic safety. All analyses are based on theoretical modeling and numerical simulation. Zemax OpticStudio 2024R1 was employed to conduct ray tracing simulations and quantify the relationship between defect characteristics and LiDAR performance degradation.

Fig. 14 shows how surface defects of wedge-shaped optical elements affect beam propagation in LiDAR systems. The diagram reveals the theoretical mechanism behind the degradation of key performance indicators such as point cloud integrity, detection range, and emergency response time. This provides a physical basis for quantitatively analyzing defect impacts on autonomous vehicle safety.

LiDAR systems determine target distance by measuring the optical signal round trip time, making them sensitive to propagation disturbances [43]. Micron-level defects on wedge prisms induce angular deviation θ , which accumulates over distance D to create displacement

$\Delta d = D \times \tan(\theta)$ at the target. These defects also attenuate laser intensity, reducing received signal power $P_{\text{returned}} \propto e^{-2\zeta Z}$. These effects cause positioning errors and missed detections, degrading LiDAR performance.

Systematic simulations studied defect sizes ranging from 3 μm to 10 μm in diameter at a distance of 10 m. Extensive ray tracing ensured data collection robustness for quantitative analysis. Key extracted parameters include lateral displacement Δd of the laser spot centroid and total optical power.

As shown in Fig. 15(a), the optical path simulation diagram illustrates the complete experimental setup, including the laser source, the wedge-shaped prism with micro-defects, and the detector on the right. Two situations are presented on the detector: the ideal spot position when there is no defect, and the offset darkened spot when there is a defect. The defect causes the spot displacement and corresponding energy loss, resulting in a darker, displaced spot. Fig. 15(b) shows the quantitative relationship curve, confirming that the increase in defect size leads to a nonlinear increase in beam deflection Δd and intensity attenuation.

Microscopic pit defects on optical components have similar impact mechanisms to hemispherical water droplets on protective covers. According to the sphere calculation formula, it can be calculated that the optical properties of a spherical pit with a diameter of 3 μm are equivalent to the influence of a 5 μL –10 μL water droplet. This equivalence enables quantitative evaluation of micro defects on LiDAR system performance indicators.

Table 3 compares the key performance indicators of the LiDAR with defective optical components (FOC) and without defective optical components (FLOC). The reliability in bad weather metric represents system performance under adverse rainy conditions based on the study by Pao et al. [35]. Other metrics include point cloud completeness, reflectivity level, detection distance, and emergency response time for comprehensive performance evaluation. Fig. 16 shows normalized metrics on a scale between 0 and 100 %, with detection distance and response time normalized using baselines of 150 m and 5.5 s.

Analysis reveals that micro defects significantly impact LiDAR performance across multiple dimensions. Point cloud completeness suffers substantial degradation, potentially causing critical obstacle information loss in long-range detection scenarios. Reflectivity reduction critically affects detection of weakly reflective targets. For high-speed driving scenarios, this translates to a meaningful loss of reaction time when every second counts for collision avoidance.

Under heavy rainfall conditions, the performance gap between defective and flawless optical components becomes even more pronounced.

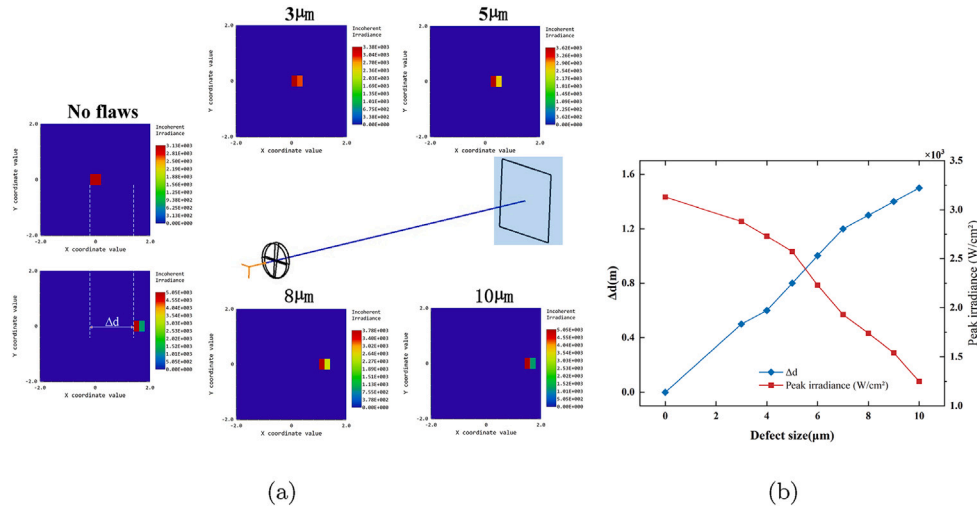


Fig. 15. Quantitative simulation defects and LiDAR performance relationship: (a) Detector status in the optical path simulation diagram, (b) Quantitative relationship between defect size and system performance degradation.

Table 3
Defects in laser lightning performance evaluation.

Performance index	FOC	FLOC	Improvement
Point cloud completeness (%)	80–90	93–97	3–17
Reflectivity level (%)	86–92	92–97	0–11
Effective detection distance (m)	136–142	146–150	4–14
Emergency response time (s)	2.2–3	1.5–2	0.2–1.5
Reliability under heavy rainfall (%)	83–90	93–98	3–15

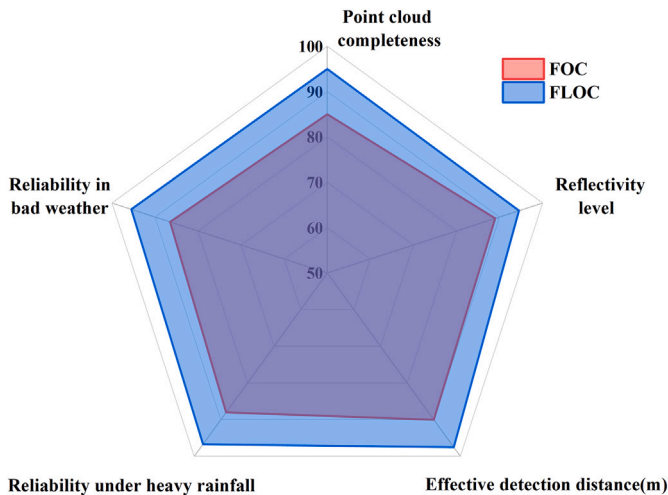


Fig. 16. Optical defect effects on LiDAR detection capabilities.

This demonstrates that manufacturing precision is crucial for maintaining system reliability under challenging environmental conditions.

In the optical component defect detection task, the imaging-optimized defect detection system can effectively identify surface defects of wedge prisms, ensuring that these optical components can guarantee the reliable operation of the entire radar system under various conditions after being installed in LiDAR, which is critical to the safety of autonomous driving. Although current laboratory conditions limit our ability to directly verify performance under specific adverse weather scenarios, such as heavy rainfall, theoretical analysis shows that micron-level surface defects can significantly reduce the detection reliability of LiDAR under these challenging environmental conditions. This result emphasizes the importance and necessity of conducting system verification research under specific severe weather conditions such as heavy rain, dense fog, and snowfall in the future.

5. Conclusion

This study successfully established a dual wavelength wedge prism imaging optimization model based on Mie scattering theory. The model significantly improved imaging quality through incident angle optimization and wavelength fusion technology. Experimental validation fully confirmed its effectiveness. Analysis confirmed that within the optimized incident angle range of 20°–40°, the dual wavelength imaging scheme is significantly superior to single wavelength methods in contrast, scattering uniformity, and imaging clarity.

A quantitative relationship model between wedge thickness and light reflection characteristics was established. This model accurately predicts spatial distribution of interference reflection and geometric reflection areas on two-dimensional image planes. Experimental verification showed high consistency with theoretical predictions, with error deviations within 20 pixels. The model provides reliable physical constraints for subsequent image processing and defect detection algorithms.

Quantitative analysis through Zemax optical simulation revealed significant impact of micron-level defects in optical components on vehicle mounted LiDAR performance. The analysis focused on performance degradation directly related to traffic safety detection distance and point cloud integrity. The proposed imaging optimization method provides important technical support for quality control of optical components during manufacturing. This ensures that defect-free components are integrated into LiDAR systems, which is significant for achieving reliable autonomous driving performance.

This study provides a comprehensive theoretical framework and laboratory validation. However, the research remains primarily theoretical. The quantitative analysis of LiDAR performance degradation is based on established optical principles and simulation results. Experimental verification in future real-world automotive environments is needed to fully confirm the practical significance of these findings. Future research should focus on experimental validation of theoretical predictions under actual severe weather conditions, such as heavy rain and fog, as well as long-term durability studies of optical components under various environmental stresses. Such comprehensive validation will bridge the gap between theoretical modeling and practical implementation, ensuring robustness and reliability of the proposed methodology in challenging operational environments.

CRedit authorship contribution statement

Jishi Zheng: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration.
Yan Wang: Writing – review & editing, Writing – original draft, Software, Data curation. **Linghua Kong:** Methodology, Formal analysis. **Dingrong Yi:** Resources, Methodology, Conceptualization. **Zhigang Ding:** Supervision, Investigation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude in order to Improve language logic. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered potential competing interests:

Jishi Zheng reports that equipment, drugs, or supplies were provided by Fujian Foctek Optoelectronics Co., Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available upon request.

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